Chemical Injection Management System – Specification

Nov 12, 2026 · Thiago Sartori Cardoso

A web platform for the chemical supplier to register tanks and wells per site, associate them under safety rules, and forecast each tank's autonomy so Sales can plan deliveries.

1. Problem framing

Sales needs a forecast, in days, of each tank's autonomy until 20% capacity, and the client needs one registry linking tanks to wells across sites.

Problem statement. The chemical supplier serves several onshore sites. Each site has tanks of different capacities and materials feeding multiple wells, and each well receives multiple products. Tank volume and well production already arrive hourly from a real-time data system, but they are not tied to an equipment registry. Sales therefore cannot see when a tank will hit the 20% reorder limit, nor how much chemical each well consumes.

Goal. Give Sales and Field Technicians one web platform to register equipment, associate tanks to wells under business rules, and forecast tank autonomy in days.

1.1 Using the "How Might We" (HMW)

- HMW: let Sales know, days in advance, which tanks will reach 20% capacity, so deliveries are planned rather than urgent?
- HMW : keep the tank–well association accurate when technicians change it in the field?
- HMW: prevent an unsafe or non-compliant chemical combination from ever being registered on a well?
- HMW : turn tank and production data into a sales argument (consumption per well, upsell opportunities)?

In scope	Out of scope (proposed, to confirm or cut)
Equipment registry: sites, tanks, wells, pumps, products	Control of pumps or injection setpoints
Tank–well association with rule validation	ERP, billing or logistics integration
Integration with the real-time data system (tank volume, well production, hourly)	Native mobile app (web only, responsive)
Injection rate per tank, consumption estimate per well, autonomy forecast	Chemical dosage engineering or recommendations
Role-based access: Sales, Field Technician, Admin	Hardware or sensor installation

In scope	Out of scope (proposed, to confirm or cut)
Multi-site support	

2. Stakeholders & personas

Sales and Field Technicians are the two primary users; the client sponsor and the real-time data system owner decide whether the project can succeed.

Stakeholder	Source	Role in the system	Need stated in the brief	Engagement (proposed)
Client sponsor (chemical supplier)	Brief	Final client; owns the system	Registry and association of tanks to wells; support to Sales	Manage closely: gate reviews
Sales representative	Brief	Only role that registers equipment; also associates	Autonomy forecast to 20%; injection rate per tank; consumption per well	Manage closely: interviews, prototype tests
Field Technician	Brief	Associates tanks to wells	Use the system without installing software	Involve: field visits, prototype tests
Real-time data system owner	Implied by the integration	Sends hourly tank volume and well production	Not stated; to confirm	Keep satisfied: technical workshop

Persona 1 – Sales Representative. Plans the next chemical deliveries and the sales potential of each site from a forecast of days until each tank reaches 20% capacity. Job to be done: *when I plan next week's route, I want to see which tanks will reach 20% first, so I can deliver on time and spot sales opportunities.*

Persona 2 – Field Technician. Works on site and has no permission to install software on the computer. Can associate tanks to wells but cannot register equipment. Job to be done: *when I reconnect a tank to a well, I want to record it quickly and be blocked if the combination is not allowed, so the data stays correct without paperwork.*

Persona 3 – Admin (proposed role, not in the brief). Manages users, sites and product master data, including the compatibility matrix.

3. OKR

Objective: Sales plans every chemical delivery from reliable, data-driven tank autonomy forecasts.

Key result	Metric	Why it matters
KR1 – Registry coverage	% of tanks registered and associated to wells with live data	Without a complete registry, forecasts are blind
KR2 – Planned deliveries	% of deliveries made before the tank reaches 20%	Measures the core promise of the brief
KR3 – Forecast accuracy	Days between predicted and actual date the tank hits 20%	Sales only trusts the tool if forecasts are right
KR4 – Adoption	% of Sales reps using the forecast weekly	Shows the tool is part of delivery planning

The brief gives no baseline or target; both are measured and agreed with the sponsor at Gate 1.

4. Assumptions, open questions & data observations

The Site 1 diagram already breaks one business rule, and the per-well consumption is an estimate, not a measurement; both must be settled with the client before detailing.

Observations from Figure 1 (Site 1)

Well	Products injected (diagram)	Note
Well-1	A, C	Violates BR "A and C cannot be applied to the same well"
Well-2	B, C + D on its flowline	D injected into the flowline, not the well
Well-3	A	—
Well-4	A, B + D on its flowline	D injected into the flowline, not the well

- Each tank has one dedicated pump (C-TK-n → Pump-n), so the pump is equipment to register too.
- Injection lines continue past Well-4, suggesting the site has more wells than shown.

Key assumptions (riskiest first)

#	Assumption	Risk if wrong	How to validate
A1	Per-well consumption can be estimated by splitting a tank's consumption in proportion to each well's production	Wrong consumption figures mislead Sales	Compare with field dosage records in discovery
A2	The real-time system sends hourly tank volume and well production through a stable API	Core features have no data	Technical workshop with the data system owner

#	Assumption	Risk if wrong	How to validate
A3	A refill can be detected as a volume increase between readings	Injection rate and autonomy spike after refills	Review sample data with refills
A4	Tanks do not serve wells on more than one site	Data model and permissions change	Confirm with Sales
A5	Field Technicians can reach a web application from their computers on site	Field association fails	Field shadowing

Check List to Open questions for the client

- ☐ Is Well-1's A + C combination real (a legacy setup to correct) or a diagram error?
- ☐ Does the 3-product limit count products injected into a well's flowline (product D)?
- ☐ Are pumps in scope as equipment, and do they report flow rate?
- ☐ Which attributes define "basic characteristics" for each equipment type?
- ☐ Does Sales need history and audit of association changes?
- ☐ Who confirms a delivery happened: the real-time data or a user?
- ☐ Are A/C the only incompatibility, or is a configurable compatibility matrix needed?

5. Project deliverables

The specification project hands over 12 deliverables; each one closes with an acceptance criterion agreed with the client sponsor.

ID	Deliverable	Phase	Format	Accepted when
D01	Project charter: problem framing, scope, OKR	Discovery	PDF	Sponsor approve the scope and OKR
D02	Stakeholder map & personas	Discovery	PDF	Validated by Sales and Field leads
D03	Discovery report: interviews, as-is process, answered open questions	Discovery	PDF	All open questions answered or logged as risks
D04	Requirements specification: functional, non-functional, business rules	Detailing	PDF / DOCX	Reviewed by Sales, Field and the data system owner
D05	Use cases & user stories with acceptance criteria	Detailing	PDF + backlog tool	Every requirement traced to at least one story
D06	Data model & calculation logic (injection rate, consumption, autonomy)	Detailing	PDF / PNG	Formulas validated on sample real data

ID	Deliverable	Phase	Format	Accepted when
D07	Integration specification with the real-time data system	Detailing	PDF	Data owner approves interface and frequency
D08	Lo-fi prototype + usability test findings	Detailing	PNG / PDF	Tested with Sales and Field users
D09	Risk register	Discovery → Planning	PDF	Every high risk has an owner and a response
D10	Suggested features & KPIs with rationale	Planning	PDF	Sponsor selects KPIs for the MVP
D11	Methodology, prioritized backlog & release plan	Planning	PDF	Dev team estimates the MVP scope
D12	README: choices, tools, feedback	Planning	MD / PDF	Submitted with the package

6. Project activities by phase

The specification runs in three phases, each closed by a gate review with the sponsor; durations are set at kickoff.

Phase 1 – Discovery

#	Activity	Technique / tool	Output
1.1	Kickoff with sponsor: goals, scope, constraints	Workshop, OKR draft	D01 draft
1.2	Map stakeholders and plan engagement	Power × interest matrix	D02
1.3	Interview Sales reps	Semi-structured interviews, JTBD	Interview notes
1.4	Shadow Field Technicians on site	Contextual inquiry	As-is process map
1.5	Technical workshop with the real-time data owner	Data sample review	Data availability notes
1.6	Validate business rules with the client	Rule review (A/C, 3-product limit)	Confirmed business rules
1.7	Consolidate findings, answer open questions, update risks	Affinity mapping	D03, D09 v1

Gate 1 – Scope confirmed: sponsor approves scope and OKR; all open questions answered or logged as risks; data access confirmed.

Phase 2 – Detailing

#	Activity	Technique / tool	Output
2.1	Write functional, non-functional requirements and business rules	Requirements catalog	D04
2.2	Write use cases and user stories with acceptance criteria	User story mapping, Gherkin	D05
2.3	Model data and define calculation logic	ER diagram, formulas tested on sample data	D06
2.4	Specify the integration interface	Interface spec (fields, frequency, failures)	D07
2.5	Build lo-fi prototype of key flows	Wireframes	D08 draft
2.6	Test prototype with users and iterate	Usability tests with Sales and Field users	D08
2.7	Review requirements with Sales, Field and the data system owner	Walkthrough	Signed-off D04–D07

Gate 2 – Requirements baselined: every requirement traces to a story; prototype validated by users; formulas validated on real data.

Phase 3 – Planning for implementation

#	Activity	Technique / tool	Output
3.1	Define KPIs and suggested features	KPI tree, business case	D10
3.2	Choose development methodology	Cynefin / uncertainty assessment	D11
3.3	Prioritize backlog and define MVP	MoSCoW + WSJF	D11
3.4	Estimate with the dev team and plan releases	Planning poker, release plan	D11
3.5	Hand over the package	Final presentation, README	D12

Gate 3 – Ready for implementation: MVP scope estimated and accepted; first sprint backlog ready (meets Definition of Ready).

7. Requirements & business rules

Every item below traces to its origin: **Brief** (the challenge text), **Diagram** (Figure 1) or **Proposed** (a specification decision to confirm with the client).

7.1 Business rules

ID	Rule	Source	Enforcement
BR-01	Only Sales can register and edit equipment	Brief	Permission
BR-02	Sales and Field Technicians can create and remove tank–well associations	Brief	Permission
BR-03	A tank can feed many wells, and a well can receive many products (many-to-many)	Brief	Data model
BR-04	A well cannot receive more than 3 chemical products	Brief	Blocks the association
BR-05	Products A and C cannot be applied to the same well	Brief	Blocks the association
BR-06	Every piece of equipment belongs to one site; the system supports many sites	Brief	Data model
BR-07	Tank autonomy is counted until the tank reaches 20% of its capacity	Brief	Calculation
BR-08	Each tank stores one product and has one pump	Diagram	Data model (to confirm)

7.2 Functional requirements

ID	Requirement	Source	Rules
FR-01	Register sites	Brief	BR-06
FR-02	Register tanks with basic characteristics, including capacity, construction material and product	Brief	BR-01, BR-08
FR-03	Register wells with basic characteristics	Brief	BR-01
FR-04	Register pumps and chemical products	Diagram	BR-01
FR-05	Associate and remove associations between tanks and wells, validating the rules before saving	Brief	BR-02 to BR-05
FR-06	Receive the hourly tank volume and individual well production from the real-time data system	Brief	—
FR-07	Show the injection rate of each tank	Brief	—

ID	Requirement	Source	Rules
FR-08	Show the estimated chemical consumption of each well, based on its individual production	Brief	BR-03
FR-09	Show each tank's autonomy in days until 20% capacity	Brief	BR-07
FR-10	Filter all views by site	Brief	BR-06

7.3 Non-functional requirements

ID	Requirement	Source
NFR-01	Web application used through a browser, with no installation on the user's computer	Brief
NFR-02	Role-based access control for Sales and Field Technician	Brief
NFR-03	Process data at the hourly frequency sent by the real-time system	Brief
NFR-04	Keep an audit trail of association changes (who, when, what)	Proposed
NFR-05	Flag tanks whose real-time data is missing or stale instead of showing a wrong forecast	Proposed

7.4 Calculation logic

All three values are calculated from the two data feeds in the brief; volumes use the unit sent by the real-time system.

Injection rate of a tank (volume per day), from the volume drop over a window of n days, excluding refills (a volume increase between readings):

```
Injection rate = (Volume n days ago − Volume now) ÷ n
```

Estimated consumption of a well fed by that tank, split in proportion to each well's production (assumption A1):

```
Well consumption = Tank injection rate × (Well production ÷ Total production of the wells fed by the tank)
```

Autonomy of a tank in days until 20% of capacity (BR-07):

```
Autonomy (days) = (Volume now − 0.20 × Capacity) ÷ Tank injection rate
```

7.5 Use cases

ID	Use case	Actor	Requirements
UC-01	Register equipment	Sales	FR-01 to FR-04
UC-02	Associate a tank to a well	Sales, Field Technician	FR-05

ID	Use case	Actor	Requirements
UC-03	Import real-time data	Real-time data system	FR-06
UC-04	Monitor tank autonomy and injection rate	Sales	FR-07, FR-09, FR-10
UC-05	Analyze consumption per well	Sales	FR-08, FR-10

7.6 User stories

ID	User story	Traces to
US-01	As Sales, I want to register a tank with its capacity, material and product, so that it can be monitored	FR-02
US-02	As Sales, I want to register wells per site, so that tanks can be associated to them	FR-01, FR-03
US-03	As a Field Technician, I want to associate a tank to a well from a browser, so that the registry matches the field	FR-05, NFR-01
US-04	As Sales, I want to see each tank's days until 20% capacity, so that I can plan the next delivery	FR-09
US-05	As Sales, I want to see each tank's injection rate, so that I understand its consumption	FR-07
US-06	As Sales, I want to see the estimated consumption per well, so that I can support sales decisions	FR-08

Acceptance criteria for US-03 (the story that enforces the safety rules):

```
Scenario: Block a fourth product on a well
  Given a well already receives 3 products
  When the user associates a tank of a fourth product to that well
  Then the system blocks the association
  And shows "A well cannot receive more than 3 products"

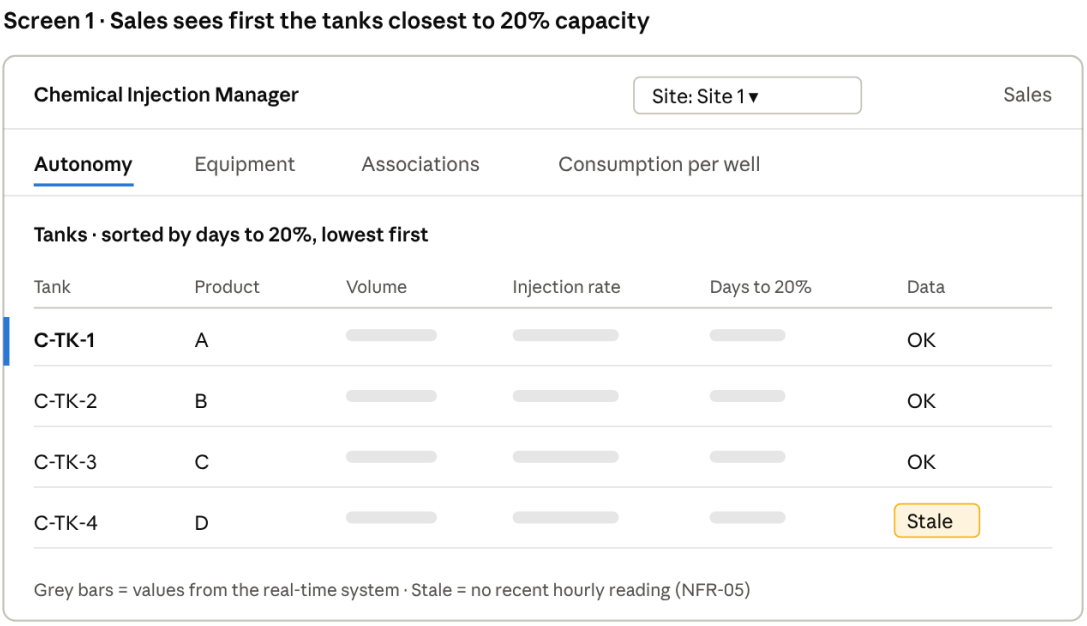
Scenario: Block products A and C on the same well
  Given Well-2 receives product C
  When the user associates a tank of product A to Well-2
  Then the system blocks the association
  And shows "Products A and C cannot be applied to the same well"

Scenario: Field Technician cannot register equipment
  Given the user is a Field Technician
  When the user opens the equipment registry
  Then the create and edit actions are not available
```

8. Lo-fi prototype

Four screens cover every user story; grey bars stand for values that come from the real-time system, and tank, well and product names come from Figure 1.

Screen 1 – Autonomy dashboard

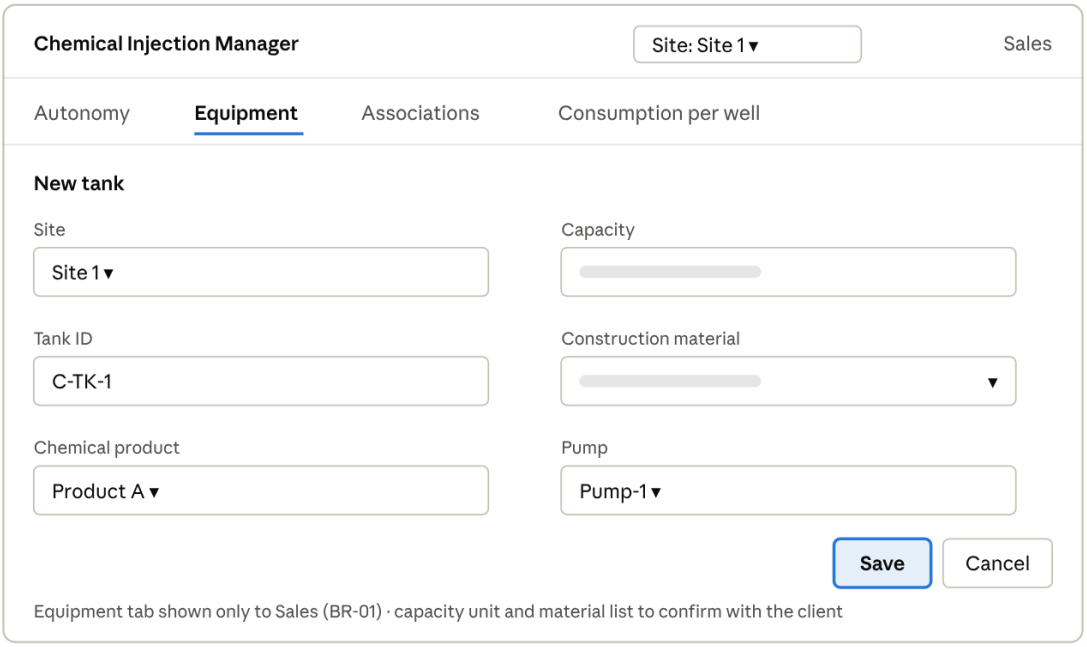


Screen 1 · Autonomy dashboard

Covers US-04, US-05 · FR-07, FR-09, FR-10, NFR-05. The site selector filters every screen; the tank closest to 20% is marked first.

Screen 2 – Tank registration

Screen 2 · Only Sales registers a tank and its basic characteristics

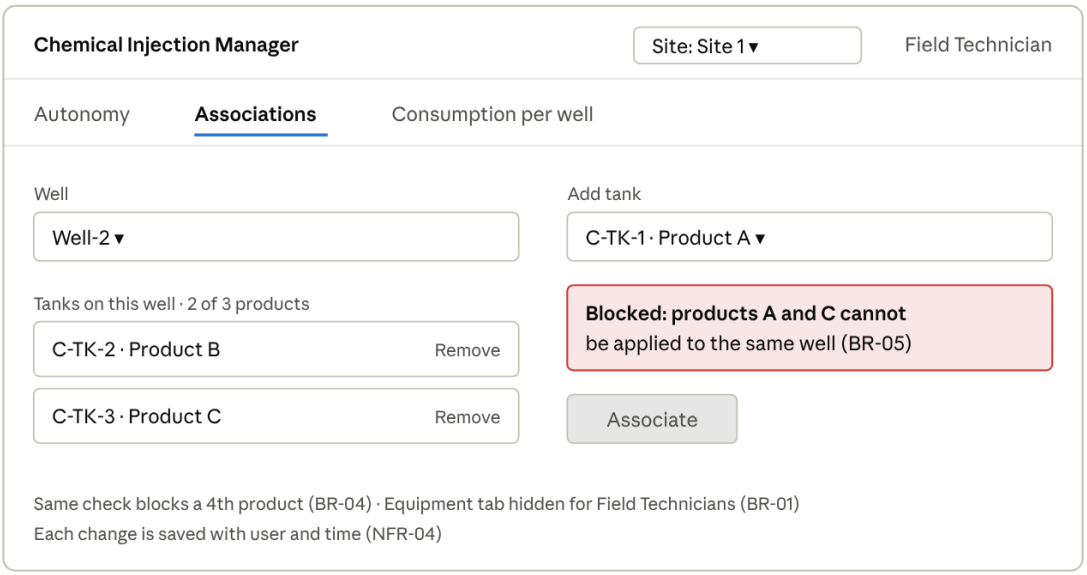


Screen 2 · Tank registration

Covers US-01 · FR-02, FR-04, BR-01, BR-08. Wells, pumps and products follow the same form pattern.

Screen 3 – Tank–well association

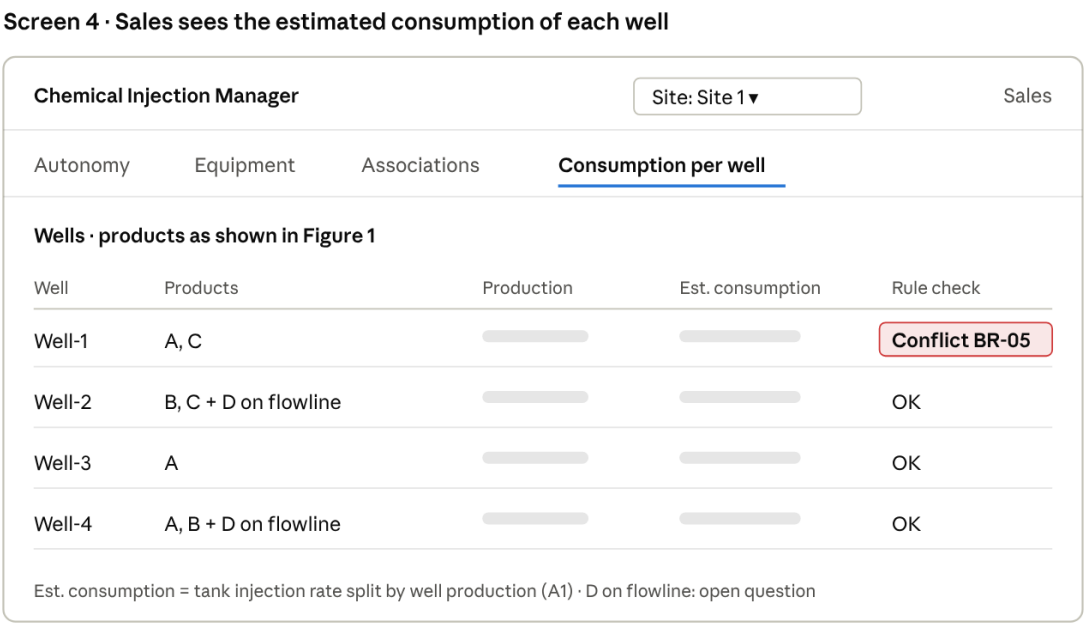
Screen 3 · The system blocks an association that breaks a rule



Screen 3 · Tank–well association

Covers US-03 · FR-05, BR-02 to BR-05, NFR-01, NFR-04. The example uses Well-2 from Figure 1, which already receives product C.

Screen 4 – Consumption per well



Screen 4 · Consumption per well

Covers US-06 · FR-08, FR-10. Product combinations come from Figure 1; the rule check surfaces Well-1's A + C conflict for the client to resolve.

9. Suggested features & KPIs

Each KPI answers a question Sales cannot answer today, and each suggested feature removes a manual step left by the brief's scope; all are **Proposed** and computed only from data the system already holds.

9.1 KPIs

KPI	Question it answers	Calculation (from system data)	Who uses it
Tanks below the reorder horizon	Which tanks need a delivery soon?	Count of tanks with days to 20% ≤ a horizon set by Sales	Sales
Days to 20% per tank	When must the next delivery happen?	Formula D (section 7.4)	Sales

KPI	Question it answers	Calculation (from system data)	Who uses it
Injection rate trend	Is a tank consuming more or less than before?	Injection rate this period vs. previous period	Sales
Chemical per unit of production per well	How much chemical does each well need for what it produces?	Est. consumption ÷ well production	Sales
Forecast accuracy	Can Sales trust the forecast?	Predicted vs. actual date each tank reached 20%	Sales, product team
Rule conflicts open	Is the registry compliant with the business rules?	Count of wells breaking BR-04 or BR-05	Sales, client
Data freshness	Are forecasts based on current data?	% of tanks with a reading in the last hour	Product team, data owner
Adoption	Is the tool part of delivery planning?	Weekly active users by role	Client sponsor

9.2 Suggested features

ID	Feature	Problem it solves	Builds on
SF-01	Alert by e-mail or in-app when a tank crosses the reorder horizon	Sales must open the dashboard to notice a critical tank	FR-09
SF-02	Delivery planning view: tanks grouped by site and due date	Sales plans routes site by site	FR-09, FR-10
SF-03	Refill detection and delivery log	A refill distorts the injection rate and is not recorded	FR-06, assumption A3
SF-04	Configurable product compatibility matrix	BR-05 is hard-coded; new products may bring new restrictions	BR-05, open question
SF-05	Association history per well	Answers "what was injected here, and since when?"	NFR-04
SF-06	Export of tables to spreadsheet / PDF	Sales shares forecasts with the client and logistics	FR-07 to FR-09

10. Development methodology & prioritization

Recommendation: build with Scrum in short sprints, organized in the SAFe hierarchy (Epic → Feature → Story) and prioritized by WSJF; the MVP is the autonomy forecast, and consumption per well follows once assumption A1 is validated.

10.1 Why an agile method

Factor in this project	Implication
Integration with an existing real-time system (A2)	Technical risk must surface early → short iterations with working increments
Consumption per well is an estimate (A1)	Formulas need validation with real data → incremental delivery and feedback
Two user roles with different needs	Frequent reviews with Sales and Field users
Requirements already baselined at Gate 2	Stable backlog to start; change is absorbed sprint by sprint

Scrum or SAFe. One delivery team → Scrum is enough. If the client already runs SAFe or the work spans several teams (e.g., the data system team), the same backlog plugs into a SAFe Agile Release Train: features are planned in PI Planning and ordered by WSJF. The method choice does not change the backlog below.

Ceremonies: sprint planning, daily, sprint review with Sales and Field users, retrospective, backlog refinement. Sprint length is proposed at 2 weeks, to confirm with the team.

Definition of Ready: story traced to an FR/BR, acceptance criteria written, dependencies known, estimated. **Definition of Done:** acceptance criteria pass, rules enforced in code, reviewed by the PO, deployed to a test environment.

10.2 MoSCoW – scope by release

Priority	Items	Reason
Must (MVP)	E1 Registry, E2 Association + rules, E3 Integration, E4 Autonomy dashboard, E6 Access control	Core need of the brief: forecast autonomy from registered, correctly associated tanks
Should (Release 2)	E5 Consumption per well, E7 Audit trail + stale-data flag	In the brief, but depends on the riskiest assumption (A1)
Could (Release 3)	SF-01 to SF-06	Proposed improvements; no requirement in the brief
Won't (this scope)	Pump control, ERP/logistics integration, native app	Out of scope (section 1)

10.3 WSJF simulation – what to build first

WSJF answers one simple question: **what brings the most value for the least effort?** The numbers below are an example to show the method. The real scores come from a session with Sales, the sponsor and the dev team.

How it works

1. Give each item a score from 1 to 13 (1, 2, 3, 5, 8, 13) on **value**, **urgency** and **risk it removes**.
2. Add the three: this is the **cost of waiting** – what we lose by not doing it now.
3. Divide by the **effort** (how big the job is). The higher the result, the sooner it goes.

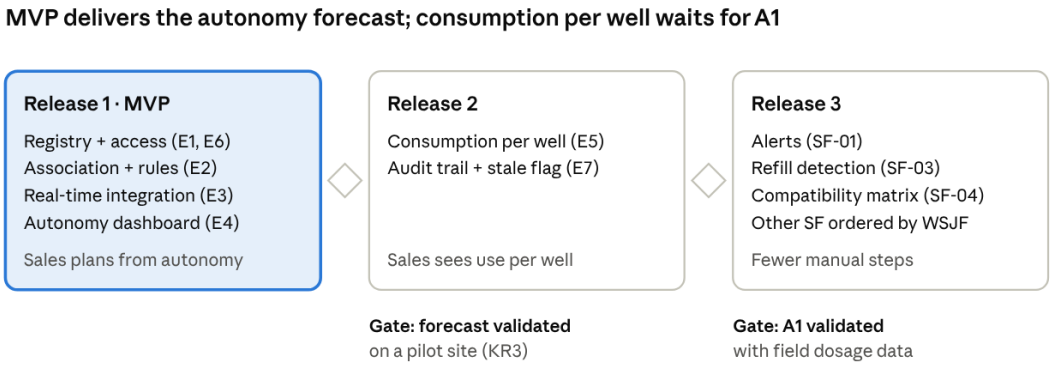
```
Priority = (Value + Urgency + Risk removed) ÷ Effort
```

Order	Item	Value	Urgency	Risk removed	Cost of waiting	Effort	Priority
1	E1 Equipment registry	5	8	8	21	3	7.0
2	E6 Access by profile	3	5	5	13	2	6.5
3	E2 Tank–well links + rules	8	8	8	24	5	4.8
4	E4 Autonomy dashboard	13	8	3	24	5	4.8
5	E7 Audit trail + stale-data flag	3	3	5	11	3	3.7
6	SF-01 Alerts	8	2	1	11	3	3.7
7	E3 Real-time data integration	8	8	13	29	8	3.6
8	E5 Consumption per well	8	3	3	14	5	2.8
9	SF-03 Refill detection	5	3	5	13	5	2.6
10	SF-04 Compatibility matrix	3	1	3	7	3	2.3

What the result tells us

- **Small jobs that unlock the rest come first:** the equipment registry and access by profile are quick and everything else depends on them.
- **The score guides, but it is not the final word:** the real-time integration (E3) has the highest cost of waiting, but it is a big job, so its score is lower. Since the dashboard cannot work without it, E3 starts in the first sprint as a technical test, alongside E1 and E6.
- **Consumption per well (E5) stays for Release 2**, in line with MoSCoW: it depends on validating assumption A1 first.

10.4 Release roadmap



Release roadmap · 3 releases, 2 gates

Each gate is a decision with the sponsor, not a date; durations come from the team's estimates and velocity after the first sprints.